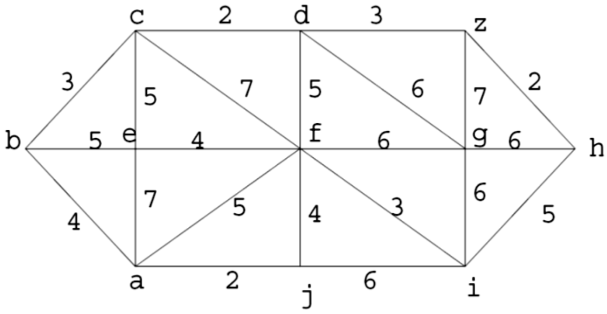
Discrete Mathematical Structure Final Exam (Spring 2013)

No :

Name:

GRAPHS

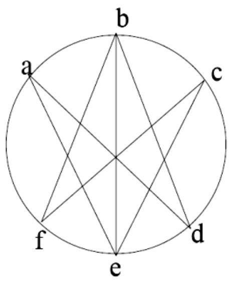
1. (15*P*) Use Dijkstra's algorithm to find the length of a shortest path and a shortest path from **a** to **z** in the following weighted graph (show each iteration in below box).



a	b	c	d	e	f	g	h	i	j	Z
0	-	-	-	-	-	-	-	-	-	-
0	4	-	-	7	5	-	-	-	2	-
0	4	7	10	7	5	11	-	8	2	-
0	4	7	9	7	5	11	13	8	2	-
0	4	7	9	7	5	11	13	8	2	**12**

The length of the shortest path is 12.

2. (10*P*) For the following graph, determine if it has an Euler path that is not a circuit.

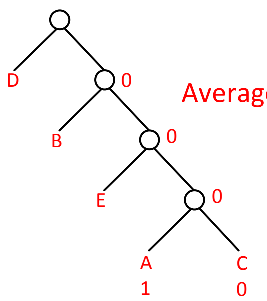


It has an Euler trail that is not a circuit because it has two vertices with odd degree (b and e).

TREES

3. (15*P*) Construct an optimal Huffman code for the set of letters in the table. Find the average length of bit strings encoding 39-letter words with the Huffman code.

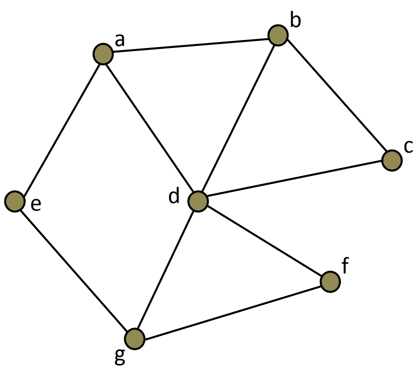
letter	frequency
A	2
B	12
C	3
D	18
E	4



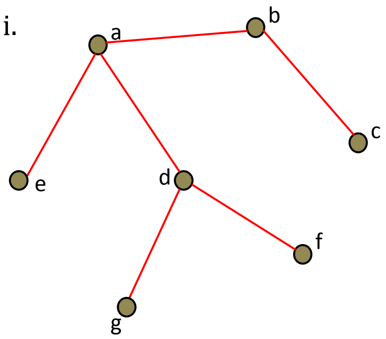
Average length $= \frac{2*4+12*2+3*4+18*1+4*3}{39} = 1.9$

A:0001

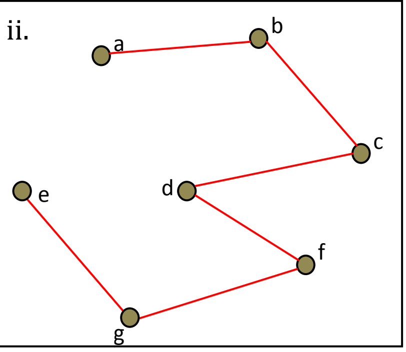
4. In the following graph with its vertices in alphabetical order find a spanning tree using
   i. (10*P*) Breadth-first search.
   ii. (10*P*) Depth-first search.



i.

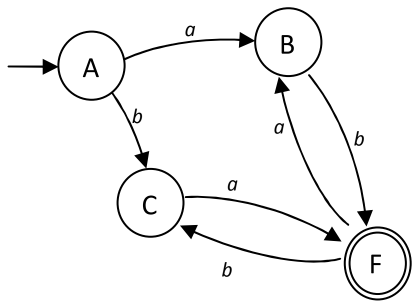


ii.



AUTOMATA

1. According to finite state automaton transition diagram given on the right,
   i. ($10P$) Design the grammar rules.
   ii. ($10P$) Describe acceptable strings as a sentence.



i.

$A \to aB$ $\quad A \to bC$ $\quad B \to bF$

$C \to aF$ $\quad F \to aB$ $\quad F \to bC$

ii.

The strings in which the number of letter "a" is equal to "b", and the same letter becomes at most twice sequential are acceptable.

2. Let G be the grammar of a language with non terminal symbols {E,F}, terminal symbol {a,b,+, *}, starting symbol E, and rules as follow.

$$E \to F \qquad E \to +EF$$

$$E \to * EF \qquad F \to a$$

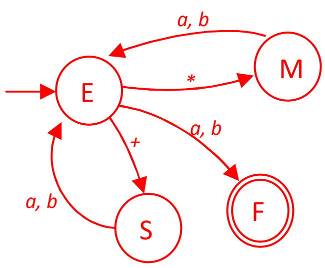
$$F \to b$$

   i. ($10P$) Find a derivation for the string "$* + a * bba$".
   ii. ($10P$) Draw its deterministic finite state automaton transition diagram by using non-deterministic one.

i.

It cannot be possible to derivate that string "$* + a * bba$" by using G grammar.

ii. (first non-deterministic one)



(then deterministic one)